

Course Number & Title: BAN 3110- Programming for Business Analytics

Semester:

Professor: Dr. Mehmet Turkoz

HOMEWORK 4

Name and ID #:_____

Instructions:

- a) You can use any A4 paper to solve the exercises. (Make sure you put the correct answers and make it readable).**
- b) You can also use word file to solve the exercises.**
- c) When you solve the problems, you can take the picture of your solutions and put these pictures on a word file if you choose option a or b.**
- d) You have to convert your word file into one pdf file. If you send more than one pdf file, your submission will not be graded. Also, do not send any file except one pdf file.**

OR You can follow following steps instead of c and d

- Take the image or scan your answer sheet
- If you use the image files, CONVERT the image files (e.g., JPG, PNG...) to a single PDF file
- If you have multiple PDFs or image files (e.g., JPG files), you can use the link
(<https://combinepdf.com/>) to combine multiple PDFs or convert multiple images to a single PDF.

-Upload one PDF file to the Blackboard "Assignments" tab

- e) Submit your solutions by Sunday by 11:59pm.**
- f) Include your pdf and py files. (You can upload these two files on Blackboard).**
- g) SUBMIT ON BLACKBOARD, NO EMAIL SUBMISSIONS**

1) [2 pts] Write a program that considers given list and prints True if all of them are odd, and False otherwise.

For example, if list L is given as L= [1, 5, 15], code will print True.

If list L is given as L= [2, 1, 5, 15], code will print False.

2) [5 pts] Write a Python code which finds the smallest even and odd numbers in a given list.

(Use for loop and if conditions for this problem. Do not use existing codes. Use your own codes).

(First find the even and odd numbers and put them in different lists (use **append** function). Then use **sort** function to find the smallest observation of each list.)

append function can be used as in the below example:

```
A=[] # We create an empty list A
```

```
# If you want to insert number 15 to the list A, You can use below code
```

```
A.append(15) # Now A=[15]
```

```
# if you also want to insert 20 to the list
```

```
A.append(20) # Now A=[15,20]
```

3) [3 pts] Write a Python code which finds common even values between given two lists A and B by using for loop and if statement? If the lists are given as follows, list1=[10,66,45,13,18,8] list2=[8,17,5,66,18,22, 6,548], then the common even values are 8,66, and 18.

4) [5 pts] How to get the **nth smallest even value** of given array? (Use for loop for this problem. Do not use existing codes. Use your own codes).

For example, in given list [11,23,58,31,56,22,43,12,65,19], if n is defined as 3. The program will print 56. (By using for loop and obtain the set of evens. By using another for loop, from the set of evens remove (n-1) observations and break the loop and find the **minimum** observation of the updated list).

You can use the remove function to **remove** the observations from a list.

remove function can be used as in the below example:

```
A=[12, 41 ,5, 16,32,54 ] # List a is given
```

```
# If you want to remove number 16 from the list A , You can use below code
```

```
A.remove(15) # Now A=[12, 41, 5, 32, 54]
```

```
# if you also want to remove 5 from the list A
```

```
A.remove(5) # Now A=[12, 41, 32, 54]
```